SFWR ENG 3S03: Software Testing

(January 7, 2025)

Note: This course outline contains important information that may affect your grade. You should retain it throughout the semester, as it will be assumed that you are familiar with the rules specified in this document.

Instructor

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Teaching Assistants

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Office hours: Tuesday, 4:30 pm to 5:20 pm, ITB 244.

Lectures

- Tuesday, Friday from 3:30 PM to 4:20 PM, in room MDCL 1102
- Wednesday from 3:30 PM to 4:20 PM, in room ABB 102

Tutorials

(T04): Alternating Wednesdays (starting Jan 22nd), 4:30-6:20, in room ABB 165

(T04): Alternating Wednesdays (starting Jan 15th), 4:30-6:20, in room ABB 165

(T04): Alternating Mondays (starting Jan 20th), 10:30-12:20, in room T13 105

Prerequisites

Basic knowledge of discrete math, data structures and algorithms. Familiar with fundamental ideas of software development. Comfortable in at least one programming language (examples will mostly make use of Java). It's helpful if you've done some testing before (and you probably have) but you don't need to have done testing systematically.

Textbooks

Recommended, nut not required (material is not taught from the textbooks)

1. Paul Ammann and Jeff Offutt. *Introduction to Software Testing*, Second Edition, Cambridge University Press, ISBN: 9781107172012.
2. Cem Kaner et al. *Lessons Learned in Software Testing: A Context Driven Approach*, Wiley, ISBN:9780471081128

Course Description

The course provides a comprehensive coverage of the fundamentals of software testing. It provides students with a foundation that can be used to assess software quality attributes and interpret measures related to software production and management. This course gives the students the opportunity to learn and practice the skills needed for a successful Software Engineering career.

Course Intended Learning Outcomes

Upon completion of this course, participants should

- Understand the fundamentals of testing & QA process, and their place within the software life cycle
- Understand the principles, scope, and best practices of testing
  - Analyse the scope and differences between various testing strategies
  - Execute error-detection techniques of inspection
  - Perform Coverage-based testing, Black-box testing, White-box testing, Debugging, Regression testing
- Understand the basic testing concepts: test cases, test plan, test report, etc.
  - Derive test cases and procedures from specifications and requirements use cases
  - Articulate test plans and test reports to document testing results
- Understand the principles of manual vs automated testing and their role in the QA process
  - Implement tests and evaluate the quality of software
  - Properly execute error-detection techniques of inspection
- Understand the fundamentals and role of measurements in Software Testing

Assessment Information

Assignments

There will be three (3) assignments. Each one of the three assignments counts for 10%. Problem sets will be handed out and will be due **around** the following dates:

Problem Set	Handed Out	Due Date
A1	January 25	February 7
A2	February 24	March 7
A3	March 24	April 6

Midterm exam

There will be one midterm exam. It will be a closed book exam. It will take place on *Tuesday, February 11*, from 3:30 PM to 4:20 PM (during the lecture time). The location of the midterm exam will be posted on the course website in due time. The mid-term exam is worth 20% of the final score.

Final exam

The final examination will be scheduled by the Registrar's office in the usual way. It will be a three hours in duration and will cover the material of the lectures, tutorials, assignments, and of the required textbook. The final exam counts for 50% of your final score.

Assessment	Weight
Assignments (3)	30%
Midterm	20%
Final Exam	50%
	TOTAL: 100%

Collaboration Policy

You may submit assignments either individually or collaboratively, under one of the following two models.

- Collaborate while analyzing the problem and developing an answer or solution, then develop the final solution independently. In this model, each person will turn in a separate document. The submissions must include a section that lists everybody you worked with and what each person contributed. You can work with as many classmates as you like with this model, but only other students in 3S03 this term
- Collaborate from start to finish with at most two other students in 3S03. You must submit one solution and each person will get the same grade. The submission must outline what each person contributed. Note: Since my experience is that students typically produce better solutions with this type of collaborative solution, I encourage students to follow this model. The incentive is a small (5%) bonus credit per assignment

Other Relevant Policy Statements

Academy Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. It is your responsibility to understand what constitutes academic dishonesty.

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the Academic Integrity Policy, located at `https://secretariat.mcmaster.ca/university-policies-procedures-guidelines/`.

The following illustrates some forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained; this includes using answers generated using ChatGPT except where you are explicitly required to use ChatGPT.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism Detection

(boilerplate text from University policy follows)

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity

For 3S03: You are expected to respect your own learning and the University's policies on academic integrity. Should you misrepresent the work you submit (e.g., by copying work done by another student or hallucinated answers generated by ChatGPT) then the standard McMaster procedures on academic integrity will be followed.

Course Modification

The instructor and university reserve the right to modify elements of the course during the term. The university may change the dates and deadlines for any or all courses in extreme circumstances. If either type of modification becomes necessary, reasonable notice and communication with the students will be given with explanation and the opportunity to comment on changes. It is the responsibility of the student to check their McMaster email and course websites weekly during the term and to note any changes.

Courses With An On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn (A2L), LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the Code of Student Rights & Responsibilities (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, whether in person or online.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, Teams or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Academic Accommodation of Students With Disabilities

Students with disabilities who require academic accommodation must contact Student Accessibility Services (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's Academic Accommodation of Students with Disabilities policy. It's helpful if you also contact me so that we can discuss the best arrangements for you, but of course you don't have to contact me if you don't want to.

Requests For Relief For Missed Academic Team Work

McMaster Student Absence Form (MSAF): In the event of an absence for medical or other reasons, students should review and follow the Academic Regulation in the Undergraduate Calendar "Requests for Relief for Missed Academic Term Work".

For 3S03: if an assignment is MSAFed then normally the value of that assignment will be combined with that of the final exam; I'm happy to discuss alternatives, for example, a short extension to the original deadline if that is appropriate. If the midterm is MSAFed then normally its value will be combined with that of the final exam. In all cases, please contact me as soon as you know you will be MSAFing a piece of work. Please keep in mind that the MSAF system sometimes doesn't send much information. Also, please double check with me sometime near the end of the term to be sure that I have recorded your MSAF correctly!

Academic Accommodation of Students With Disabilities

Students who require academic accommodation must contact Student Accessibility Services (SAS) to make arrangements with a Program Coordinator. Academic accommodations must be arranged for each term of study. Student Accessibility Services can be contacted by phone 905-525-9140 ext. 28652 or e-mail sas@mcmaster.ca. For further information, consult McMaster University's Policy for Academic Accommodation of Students with Disabilities.

Academic Accommodation For Religious, Indigenous Or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, indigenous or spiritual observances should follow the procedures set out in the RISO policy. Students should submit their request to their Faculty Office normally within 10 working days of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests. The instructor understands that sometimes accommodations and requests for alternative arrangements have to come in very late, but please try to help the instructor help you by submitting requests ASAP.

Copyright And Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, including lectures by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, A2L and/or McMaster email.